CLASS 10 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Assertion (A): Acquired traits are not inherited by the next generation. Reason (R): Changes in body cells do not change the DNA of germ cells. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. In photosynthesis, six molecules of carbon dioxide combine with water to form how many molecules of glucose? 1

(a) 2 **(b)** 12 **(c)** 1 **(d)** 6

3. A student touches a hot object, withdraws the hand and then feels the heat. The reason is: 1

(a) motor neuron is slower
(b) no relay neuron
(c) reflex action is faster than brain sensation
(d) brain acts before spinal cord

4. Rhizopus is grown on bread in a sealed, completely dry, sterile packet. Growth is absent because 1

(a) spores are not produced **(b)** bread kills them
(c) sporangia are absent **(d)** spores need moisture to germinate

5. In a cross Tt x Tt, three offspring are tall. The probability that the fourth is dwarf is 1

(a) 3/4 **(b)** 1/2 **(c)** 1/4 **(d)** 1

6. A leaf variegated with green and white patches shows starch only in green regions because: 1

(a) starch is stored there **(b)** chlorophyll is needed for photosynthesis
(c) white parts get no light **(d)** white parts are dead

7. A woman has swollen neck and slow metabolism. Her diet is deficient in: 1

(a) calcium **(b)** iodine **(c)** zinc **(d)** iron

8. If the placenta fails to form properly, the main problem for the embryo is 1

(a) shortage of nutrients and oxygen **(b)** lack of DNA
(c) excess heat **(d)** lack of chromosomes

9. In pea plants F2 seeds show round-green and wrinkled-yellow types that were absent in both parents. This happens due to 1
(a) loss of genes in F1
(b) new combinations of independently inherited genes
(c) acquired traits
(d) blending of traits

10. Assertion (A): Bile juice is necessary for digestion of fats. Reason (R): Bile contains the enzyme lipase that digests fats. 1
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. Assertion (A): Plant roots show positive geotropism. Reason (R): Roots grow upwards against gravity. 1
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

12. Assertion (A): Testes of mammals lie outside the abdominal cavity. Reason (R): Sperm formation requires a lower temperature than the body temperature. 1
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

13. Assertion (A): A recessive trait is not seen in the F1 generation. Reason (R): The dominant allele masks the recessive allele in heterozygotes. 1
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. A heart beats 72 times per minute. The number of beats in 1 hour is: 1
(a) 720 **(b)** 4320 **(c)** 432 **(d)** 7200

15. An impulse is blocked in the motor neuron only. The patient will: 1
(a) move but not feel **(b)** feel the touch but not move the muscle
(c) neither feel nor move **(d)** feel more

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why is the hormone from a gland not released continuously in the same amount? 2

17. In which situation would a plant need a pollinator the most: a unisexual flower or a bisexual flower? Give a reason. 2

18. Explain the reason a plant with 'Tt' genotype shows only the tall phenotype. 2

19. If a person has anaemia, why does he feel tired? 2

20. Why do we sweat and breathe fast when frightened? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. How many chromosomes does each of the following have in humans: sperm, egg, zygote, liver cell? What is the number of autosomes in each? 3

22. Describe how digestion of starch occurs from the mouth to the small intestine. 3

23. Differentiate between the effects of adrenaline and insulin on blood sugar. 3

24. What would happen if the uterine lining did not thicken each month? 3

OR

Explain why every egg of a woman carries an X chromosome, though she has two X chromosomes. 3

25. Explain why a person who is climbing high mountains breathes faster and has a higher RBC count. 3

OR

Explain how the nervous system and hormones act together when we see a snake. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Compare ATP production in aerobic respiration, anaerobic in yeast and anaerobic in muscle, with the sites and products and the reason for the energy difference. 5

OR

A pea seedling is placed horizontally. Predict the growth of the root and shoot after a few days, explain the mechanism including auxin. 5

27. Explain the sequence of events that happen in a human female from the release of an egg to the formation of the placenta. 5

OR

Two pea plants, both tall, produce offspring of which one-fourth are dwarf. A third tall plant crossed with a dwarf plant gives all tall offspring. Deduce the genotype of each of the three tall plants and justify. 5

END OF QUESTION PAPER Total: 50 Marks

MASTERMIND